

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 10)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station

☐

B Tropospheric ozone

☐

C Carbon monoxide from a car exhaust

☐

D Nitrogen monoxide from a diesel engine

☐

0 1 . 2

Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 2 . 1

Complete Table 1 to show the source and disposal method for each category of radioactive waste from nuclear electricity generation.

[6 marks]

Table 1

Waste category	Example / source	Storage or disposal method	
High-level waste			
Intermediate-level waste			
Low-level solid waste			

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0 2 . 2

Explain why high-level waste must be cooled and why its storage must be monitored for very long periods.

[2 marks]

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0 2 . 3

Describe how low-level liquid and gaseous wastes are treated before discharge.

[2 marks]

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0 3

Table 1 shows information about some greenhouse gases.

Table 1

Gas	Main anthropogenic sources	Approximate residence time	Global warming potential (per molecule, relative to CO₂ over 100 years)
Carbon dioxide	Fossil fuel combustion, deforestation, cement	100+ years (variable)	1
Methane	Rice paddies, cattle, landfill, gas leaks, coal mines	About 12 years	28
Nitrous oxide (NO _x)	Fertilisers, combustion	About 120 years	265
CFCs	Refrigerants, aerosols (now banned)	50-100+ years	5 000-10 000
Tropospheric ozone	Secondary pollutant from NO _x and hydrocarbons	Days-weeks	-

0 3**. 1**

Explain the natural greenhouse effect.

[3 marks]

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0 3**. 2**

Use Table 1 to explain why carbon dioxide is considered the most important anthropogenic greenhouse gas even though methane and CFCs absorb far more infrared per molecule.

[3 marks]

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0 3 . 3

Explain why reducing methane emissions would produce a faster reduction in warming than reducing CO₂ emissions by the same mass.

[2 marks]

0 3 . 4

Give **two** methods of reducing methane emissions from agriculture or waste.

[2 marks]

1

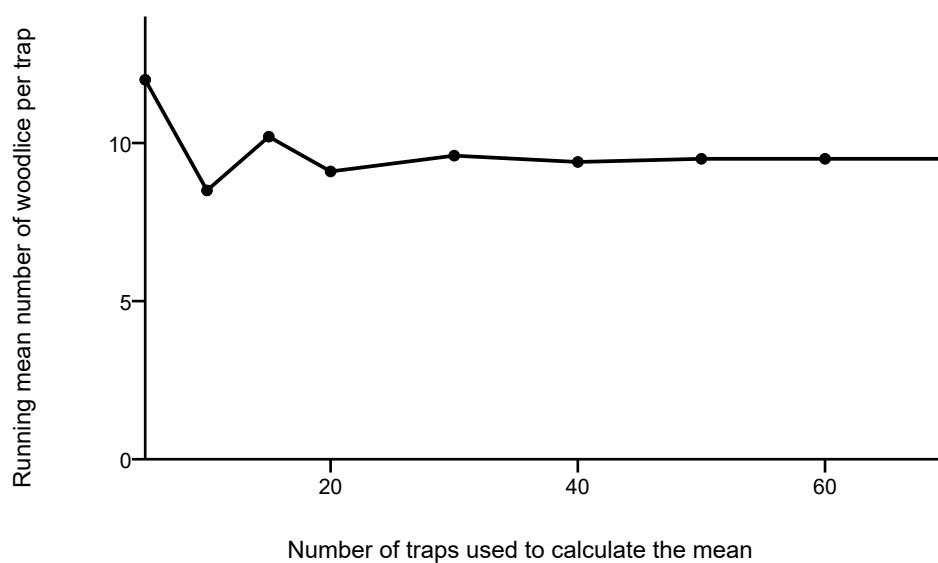
2

10

0 4

Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



04 . 1

Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

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04 . 2

Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

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04 . 3

The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

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04 . 4

Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

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10

0 5 A pumped-storage HEP station has an upper reservoir 500 m above its turbines. It can hold 7 million m^3 of water. The density of water is 1000 kg m^{-3} and $g = 9.8 \text{ N kg}^{-1}$.

0 5 . **1** Describe how pumped-storage HEP is used for peak shaving.

[3 marks]

0 5 . **2** Calculate the maximum gravitational potential energy stored when the upper reservoir is full. Give your answer in standard form.

[3 marks]

Answer

J

0 5 . **3** The station returns 75% of the electricity used for pumping. Explain why the process is still worthwhile despite this loss.

[2 marks]

0 5 . 4

Give **two** locational requirements of a pumped-storage station.

[2 marks]

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0 6

A dam and reservoir are to be built on a river to supply water and regulate flow.

0 6 . 1

Explain **three** effects the reservoir may have on wildlife in and around the river.

[6 marks]

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0 6 . 2

Explain why decomposition of flooded vegetation in a new reservoir contributes to climate change.

[2 marks]

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06 . 3

Explain how a river-regulation reservoir reduces both flooding and water shortages downstream.

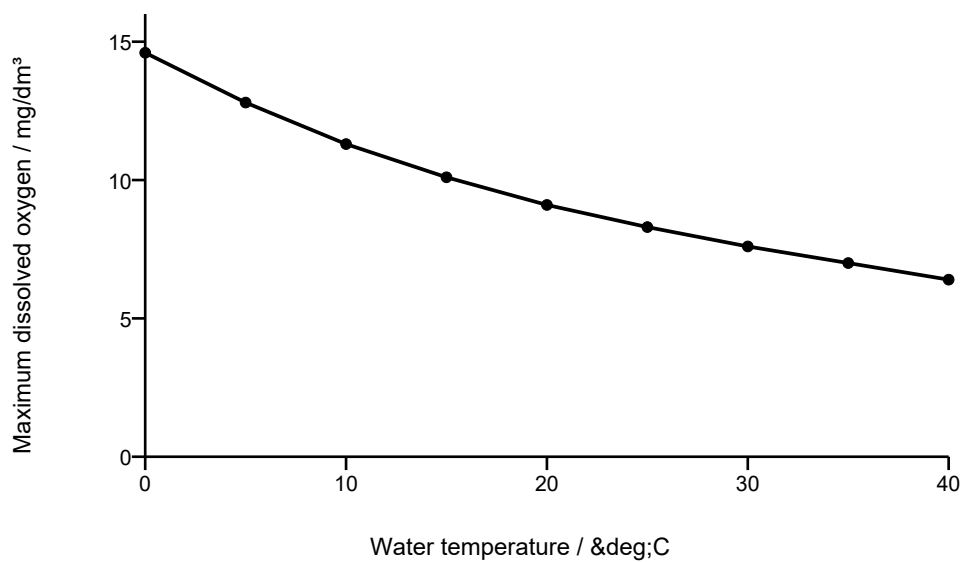
[2 marks]

10

07

Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg dm^{-3} .

Figure 1



07 . 1

Describe the relationship shown in Figure 1.

[1 mark]

07 . 2

Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

[1 mark]

Answer

°C

07 . 3

In summer the river upstream of the power station is at 20 °C and the effluent raises the temperature by 7 °C. Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

[2 marks]

Answer

mg/dm³

07 . 4

Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1.

[2 marks]

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07 . 5

Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen.

[3 marks]

1

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0 7 . 6

Describe how a cooling tower reduces the temperature of power station effluent water.

[1 mark]

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10

0 8 . 1

Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

[3 marks]

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0 8 . 2

Explain the environmental impacts of mining polymetallic nodules.

[3 marks]

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08 . 3

Explain why rare earth metals are a supply concern and describe one new extraction technique.

[4 marks]

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09 . 1

Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction.

[3 marks]

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09 . 2

Describe **four** environmental concerns about fracking and suggest how they may be reduced.

[5 marks]

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09 . 3

Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

[2 marks]

10

10 . 1

Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

[illegible]

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10 . **2** State what is meant by the **Leq 57 dB** contour used around UK airports. **[1 mark]**

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10

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Write an essay on one of the following topics.

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Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

OR

1 1 . 2

Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

END OF QUESTIONS